

QuantOS - AI Model Training: Critical Opinion

What to remove, add, or replace - and why - with a focus on the model-training pipeline

Executive summary

QuantOS is a serious, well-governed research-and-backtesting platform for NSE India equity and derivatives. Every invariant is real - exact Decimal accounting, no-lookahead next-bar fills, real dated STT/GST/exchange costs, immutable content-addressed evidence, and multiplicity-deflated Sharpe. The engineering is rare and trustworthy. But trustworthiness is not profitability. After 101 governed trials across two campaigns plus six research screens, the best candidate reached a deflated Sharpe of 0.398 against a 0.95 promotion gate - less than half the bar. The honest evidence says the model and features are the binding constraint, not the governance.

1. What the system currently does (model training today)

The ML stack lives in `src/quant_system/modeling/`. The canonical pipeline is:

- **Features:** a 400-bar canonical window producing a fixed six-feature vector (built in `mizan_features.py` / `features.py`).
- **Labels:** next-open-to-following-open net-cost label, with purged/embargoed folds to kill lookahead.
- **Model family:** ridge regression - a single linear model, with four timing-/cost-aligned baselines.
- **Validation:** expanding walk-forward fold, deflated Sharpe, sampling-aware deflation for the full search count.
- **Promotion gate:** deflated Sharpe > 0.95 plus stress tests, governance checks, and an evidence-only verdict.
- **Persistence:** every trial immutably recorded with content-hashed prediction/metric hashes.

1a. What the evidence shows

Two governed campaigns (101 trials), six research screens, and six pre-declared research screens found **no edge that survives real costs**. The best deflated Sharpe was 0.398. The platform records this rather than hiding it - which is a feature of the engineering, but also a verdict on the current model/feature combination.

2. REMOVE / DEPRECATE

2.1 Honesty overhead on the hot path

The content-addressed evidence store, lease-based access, and publication machinery are excellent for governance, but they add latency and complexity that the *training* hot path does not need. Move them to a governance sidecar; let the training step import a lean feature vector and write a raw checkpoint, then have the evidence layer wrap it post-hoc. Right now the publication step is entangled with the training loop.

2.2 Three parallel feature-derivation paths

`alpha/technical.py`, `modeling/mizan_features.py`, and `execution/mizan_live_features.py` all derive features from the same bars. This is a maintenance tax and a source of subtle lookahead bugs - a value computed in one module with one purge convention and a slightly different convention in another. **Consolidate to a single canonical feature pipeline** that everyone imports from.

2.3 The assumption that more folds = more learning

If the features and model family cannot express an edge, running 1,000 folds just confirms that precisely - and burns compute. Add an early **feature-utility gate**: compute out-of-sample IC for every candidate feature; kill any below threshold before it enters the fold pipeline. The current system seems to keep feeding weak features into ridge and celebrating the rigor of the process.

3. ADD

3.1 Alternative model families (the single biggest gap)

The platform commits to ridge regression. Ridge is interpretable and well-regularized, but markets are nonlinear and regime-dependent. Ridge cannot capture threshold effects, regime switches, or interactions between, say, IV surface curvature and order-flow imbalance. **Add at least one nonlinear family:**

- Gradient-boosted trees (XGBoost / LightGBM) for tabular feature sets - handles nonlinearities and interactions natively.
- A small temporal model (1D-CNN or Transformer encoder) for sequential patterns in the 400-bar window.
- Let the promotion gate *compare* families on the same fold and promote the one that clears the deflated-Sharpe bar, not the one that is easiest to implement.

3.2 Feature importance and selection infrastructure

When 101 trials have found no edge, the first question is *are the features wrong?* not *is the training pipeline correct?* Add permutation importance / SHAP, a recursive feature elimination step, and an out-of-sample IC screen that prunes dead features before they pollute the fold. The current *mizan_features.py* probably carries dozens of candidate features - knowing which ones are dead is more valuable than another fold of the same set.

3.3 Alternative / alternative data surfaces

The README explicitly excludes fundamentals and sentiment. That is a defensible scope boundary, but it is also why 101 trials found nothing. Price + volume + Greeks on NSE is a crowded feature space. For a different edge you need:

- **Order-flow imbalance** - NSE provides tick-by-tick trades; a simple imbalance signal is one of the strongest documented intraday edges in Indian markets.
- **Sentiment** - India-specific financial media and options-flow commentary (the repo already has *scripts/ingest_macro_regimes.py* but it does not feed the model).
- **Macro regimes** - the macro ingestion script exists but is not wired into *mizan_features*. Connecting it is a low-hanging, high-leverage change.

3.4 Regime-adaptive windows

The canonical 400-bar window is fixed. Markets have regimes (trending, mean-reverting, volatile, calm) and a window optimal in one regime is suboptimal in another. Add a regime detector (even a simple volatility-regime switch) and let the feature window adapt. This is a small architectural change with potentially large marginal impact and is more likely to surface an edge than another ridge fit on a fixed window.

3.5 Ensemble / stacking

Once multiple model families exist, stack them: ridge for linear structure, trees for nonlinear interactions, a neural net for sequential patterns, and a simple logistic or ridge meta-learner on their out-of-sample predictions. The current promotion pipeline evaluates models in isolation; stacking often beats any single model.

4. REPLACE

4.1 Ridge to nonlinear model family as default

Replace ridge as the primary candidate with gradient-boosted trees. Keep ridge as a baseline - it is a good sanity check - but promote the model that clears the deflated-Sharpe gate, not the one that is easiest to implement. The fact that the whole system is named Mizan and the best Sharpe is 0.398 suggests the **model representation is the binding constraint**, not the training loop.

4.2 Fixed 400-bar window to adaptive multi-scale features

Replace the single canonical window with a multi-scale feature set (short / medium / long horizons) plus a regime indicator that selects which scale matters. This is a small architectural change with large marginal impact.

4.3 Compute-everything to disciplined IC-screen-then-rank

Replace the current philosophy of computing everything in the 400-bar window with a disciplined pipeline: compute ~100 candidate features, screen each for out-of-sample IC stability, keep only those above threshold, feed the survivors to the model. The current pipeline probably carries a lot of redundant, correlated, or weak features diluting the signal.

5. WHY - THE UNDERLYING ARGUMENT

QuantOS is engineered to be **right** about what it computes, but not engineered to **find an edge**. That is the crucial distinction. Every invariant - no lookahead, exact Decimal, deflated Sharpe, multiplicity - makes the result trustworthy. But trustworthiness is not profitability.

The binding constraint is almost certainly **the model + features, not the governance**. After 101 trials with ridge and a fixed 400-bar window on price/volume/Greeks and no edge, the honest conclusions are:

- The feature space is not expressing what is actually tradeable, *or*
- The alpha has already been arbitrated away on NSE cash equities, *or*
- The edge lives in a different instrument / data type (options flow, sentiment, macro regime) that the system explicitly excluded.

The highest-expected-value moves are therefore: **(1) add nonlinear models, (2) add alternative data, (3) add feature selection, (4) make the window adaptive**. Everything else is optimization of a system that has not yet found something worth optimizing.

6. RECOMMENDED PRIORITY ORDER

#	Change	Rationale	Effort
1	Add gradient-boosted trees (XGBoost/LightGBM)	Ridge is linear; markets are nonlinear. This alone may unlock the edge.	Medium
2	Wire macro regimes into mizan_features	Script already exists; just not connected. Regime-aware features are low-hanging fruit.	Low
3	Add permutation importance / recursive feature elimination	Answers 'are the features wrong?' before 'is the training correct?'.	Medium
4	Multi-scale adaptive windows + regime switch	Fixed 400-bar window is a hard constraint; adaptive scales are more flexible.	Medium
5	Add order-flow imbalance features	Strongest documented intraday edge in NSE; requires tick data pipeline.	High
6	Ensemble stacking of multiple families	Only useful after (1)-(4) give you multiple competing model outputs.	Medium
7	Move evidence-store machinery off the training hot path	Reduces latency and complexity in the training loop; governance stays intact.	Medium

Bottom line: QuantOS is well-built and its honesty is a genuine asset, but the empirical signal it has produced is that NSE markets, after real costs and proper deflation, have not yielded a profitable edge across 101 trials with ridge on a fixed window. The next phase should optimize for *expressiveness* (nonlinear models, alternative data, adaptive features) rather than further tightening a governance layer that is already airtight.

